

CLASS 8 · CBSE · GANITA PRAKASH PART 2 · SHORT NOTES

## Chapter 3: Proportional Reasoning – 2

Topic-wise short notes with solved examples

Map scales, ratios with more than two terms, dividing a whole into several parts, pie charts, inverse proportion, and problems on work done together.

### 1 Recap of proportion

- Two quantities are in **direct proportion** when they change by the same factor. Their quotient stays constant:  $\frac{x_1}{y_1} = \frac{x_2}{y_2} = k$ .
- $a : b :: c : d$  exactly when  $a \times d = b \times c$ .
- Rule of Three: if  $a : b :: c : d$  then  $d = \frac{bc}{a}$ .

**Example 1.1** One cook mixes 6 cups of rice with 3 cups of dal, and another mixes 4 cups of rice with 2 cups of dal. Are the mixtures the same?

SOLUTION

$$6 \times 2 = 12 \text{ and } 3 \times 4 = 12$$

**Yes, both are 2 : 1**

### 2 Ratios in maps

- The **Representative Fraction (RF)** of a map is the ratio of a distance on the map to the actual distance on the ground, in the same unit.
- RF 1 : 60,00,000 means 1 cm on the map stands for 60,00,000 cm, which is 60 km.
- Useful conversion: 1 km = 1000 m = 1,00,000 cm.
- A map gives the straight-line distance, not the road distance.

**Example 2.1** A map has RF 1 : 25,00,000. Two towns are 6.4 cm apart on the map. Find the actual distance.

SOLUTION

$$1 \text{ cm stands for } 25,00,000 \text{ cm} = 25 \text{ km.}$$

$$6.4 \times 25$$

**160 km**

**Example 2.2** A classroom 8 m long is drawn to the scale 1 : 50. How long is it in the drawing?

SOLUTION

$$800 \text{ cm} \div 50$$

**16 cm**

### 3 Ratios with more than two terms

- A ratio  $a : b : c : d$  says that for every  $a$  units of the first quantity there are  $b$  of the second,  $c$  of the third and  $d$  of the fourth.
- Two such ratios are proportional,  $a : b : c : d :: p : q : r : s$ , when  $\frac{a}{p} = \frac{b}{q} = \frac{c}{r} = \frac{d}{s}$ .
- Simplest form: divide every term by the HCF of all the terms.
- If one term is known, find the value of **one part** and multiply.

**Example 3.1** A spice mix uses coriander, chillies, dal and fenugreek in the ratio 8 : 4 : 2 : 1. If only 2 chillies are available, how much of each of the others is needed?

SOLUTION

Chillies drop from 4 to 2, a factor of  $\frac{1}{2}$ . Apply it to every term.

**4 spoons of coriander, 1 spoon of dal and half a spoon of fenugreek**

**Example 3.2** Red, blue and white paint are mixed in the ratio 2 : 3 : 5. With 10 litres of white, how much red and blue are needed, and how much paint is made?

SOLUTION

5 parts = 10 litres, so 1 part = 2 litres.

**4 litres of red and 6 litres of blue; 20 litres in all**

**Example 3.3** A library has Odia, Hindi and English books in the ratio 3 : 2 : 1. There are 288 Odia books. Find the others.

SOLUTION

3 parts = 288, so 1 part = 96.

**192 Hindi books and 96 English books**

### 4 Dividing a whole in a given ratio

- To divide  $x$  in the ratio  $a : b : c : \dots$ , the parts are

$$\frac{a}{a + b + c + \dots} \times x, \quad \frac{b}{a + b + c + \dots} \times x, \quad \frac{c}{a + b + c + \dots} \times x, \dots$$

**Example 4.1** Concrete is made from cement, sand and gravel in the ratio 1 : 1.5 : 3. How much of each is needed for 110 units of concrete?

SOLUTION

Total parts = 5.5, and  $110 \div 5.5 = 20$ .

**20 units of cement, 30 of sand and 60 of gravel**

**Example 4.2** The angles of a triangle are in the ratio 1 : 3 : 5. Find them.

SOLUTION

The sum is  $180^\circ$  and there are 9 parts, so one part is  $20^\circ$ .

**$20^\circ$ ,  $60^\circ$ ,  $100^\circ$**

**Example 4.3** A 150-minute practice session is split among warm-up, batting, bowling and fielding in the ratio 3 : 4 : 3 : 5. Find the time for each.

SOLUTION

Total parts = 15; one part = 10 minutes.

**30, 40, 30 and 50 minutes**

**Example 4.4** There are 100 coins of ₹10, ₹5, ₹2 and ₹1 in the ratio 4 : 3 : 2 : 1. Find the total value.

SOLUTION

The numbers of coins are 40, 30, 20 and 10.

Value =  $400 + 150 + 40 + 10$

**₹600**

**Example 4.5** Can a triangle have sides in the ratio 1 : 3 : 5?

SOLUTION

The sides would be  $k$ ,  $3k$ ,  $5k$ , and  $k + 3k = 4k$  is less than  $5k$ .

**No. The sum of two sides must be greater than the third.**

## 5 Pie charts

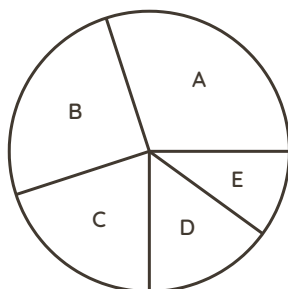
- A **pie chart** shows how a whole is divided into parts. Each part is a sector (slice) of a circle.
- The angle of a sector is proportional to the quantity it stands for. The angles add up to  $360^\circ$ .

$$\text{Angle of a sector} = \frac{\text{value of the part}}{\text{total value}} \times 360^\circ$$

- If the data are percentages: angle = percentage  $\times 3.6^\circ$ .

- **To draw:** draw a circle and one radius; measure the first angle from it and draw the next radius; continue from each new radius; label the sectors.

| Grade    | A           | B          | C          | D          | E          | Total       |
|----------|-------------|------------|------------|------------|------------|-------------|
| Students | 12          | 10         | 8          | 6          | 4          | 40          |
| Angle    | $108^\circ$ | $90^\circ$ | $72^\circ$ | $54^\circ$ | $36^\circ$ | $360^\circ$ |



Pie chart of grades: each student is  $9^\circ$

**Example 5.1** 360 people chose a favourite season: 90 summer, 120 rainy and the rest winter. Find the angles for a pie chart.

**SOLUTION**

Winter =  $360 - 210 = 150$  people. Here each person is  $1^\circ$ .

**Summer  $90^\circ$ , rainy  $120^\circ$ , winter  $150^\circ$**

**Example 5.2** Favourite TV channels: entertainment 50%, sports 25%, news 15%, information 10%. Find the angles.

**SOLUTION**

Multiply each percentage by  $3.6^\circ$ .

**$180^\circ, 90^\circ, 54^\circ, 36^\circ$**

**Example 5.3** In a pie chart the sector for "car" has an angle of  $60^\circ$  and stands for 18 children. How many children were surveyed? How many does a  $90^\circ$  sector stand for?

**SOLUTION**

$60^\circ$  is  $\frac{1}{6}$  of the circle, so the total is  $18 \times 6 = 108$ .

$90^\circ$  is  $\frac{1}{4}$  of 108.

**108 children; 27 children**

## 6 Inverse proportion

- Two quantities are in **inverse proportion** when one is multiplied by a factor  $n$  and the other is multiplied by  $\frac{1}{n}$ . Their **product stays constant**:  $xy = k$ .
- If  $(x_1, y_1)$  and  $(x_2, y_2)$  are two pairs of values, then  $x_1y_1 = x_2y_2$ , and so  $\frac{x_1}{x_2} = \frac{y_2}{y_1}$ .

|                     |    |    |    |     |
|---------------------|----|----|----|-----|
| Speed (km/h)        | 5  | 15 | 30 | 60  |
| Time (hours)        | 18 | 6  | 3  | 1.5 |
| Speed $\times$ time | 90 | 90 | 90 | 90  |

- Examples: speed and time for a fixed distance; number of workers and days for a fixed job; number of pumps and time to fill a tank; number of people and days a stock of food lasts.
- Direct or inverse?** Ask: if one quantity doubles, does the other double (direct) or halve (inverse)?
- These problems assume that every worker, pump or machine works at the same steady rate.

**Example 6.1** Are  $x$  and  $y$  in inverse proportion?  $x$ : 40, 80, 25, 16 and  $y$ : 20, 10, 32, 50.

SOLUTION

The products are 800, 800, 800, 800.

**Yes**

**Example 6.2** 20 workers lay a road in 4 days. How long will 10 workers take?

SOLUTION

$$20 \times 4 = 10 \times d$$

**8 days**

**Example 6.3** 2 pumps fill a tank in 18 hours. How long do 4 pumps take?

SOLUTION

$$2 \times 18 = 4 \times t$$

**9 hours**

**Example 6.4** Food for 80 students lasts 15 days. If 20 more students join, how long will it last?

SOLUTION

$$80 \times 15 = 100 \times d$$

**12 days**

**Example 6.5** 42 machines make a batch of toys in 63 days. How many machines are needed to make it in 54 days?

SOLUTION

$$42 \times 63 = m \times 54$$

**49 machines**

**Example 6.6** A car takes 2 hours at 60 km/h. How long does it take at 80 km/h?

SOLUTION

$$60 \times 2 = 80 \times t$$

**1.5 hours**

## 7 Working together

- Take the whole job as **1 unit of work**. If a person finishes it in  $t$  hours, the work done in 1 hour is  $\frac{1}{t}$ .
- Add the rates of all the workers to get the work done together in 1 hour. The time taken together is the reciprocal of that sum.
- If two workers take  $a$  and  $b$  hours alone, together they take  $\frac{ab}{a+b}$  hours.

**Example 7.1** Ram cuts a basket of vegetables in 1 hour and Shyam takes 1.5 hours. How long do they take together?

SOLUTION

Rates: 1 and  $\frac{1}{1.5} = \frac{2}{3}$  of the job per hour. Together:  $\frac{5}{3}$  per hour.

**$\frac{3}{5}$  hour, that is 36 minutes**

**Example 7.2** A small pump fills a tank in 3 hours and a large pump in 2 hours. How long do both take together?

SOLUTION

Rates:  $\frac{1}{3} + \frac{1}{2} = \frac{5}{6}$  of the tank per hour

**$\frac{6}{5}$  hours, that is 1 hour 12 minutes**

## 8 Common mistakes

- Using the Rule of Three for an inverse proportion. More workers mean **fewer** days.
- Treating every "more means more" situation as a proportion. Check that the factor is the same.
- Forgetting to change centimetres to kilometres when reading a map.
- Dividing by the number of terms instead of the **sum** of the terms when sharing in a ratio.
- Drawing pie-chart angles that do not add up to  $360^\circ$ .
- Adding the times when two people work together. Add the **rates**.

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